

Grey Matter: Following What Changes in Lived Experience

Human-readable companion to Grey Matter: Where What's Real Gets Shaped Into What's Lived - A Perturbation-Response Architecture for Conscious Experience

Human-readable companion to the completed scientific manuscript. This version preserves the scientific claims and limits of Grey Matter while presenting the motivating problem and inferential architecture in more direct language. It does not constitute a separate research contribution and does not replace the authoritative manuscript.

Abstract

A person's conscious life is continually exposed to changing sensory input, bodily signals, memories, expectations, goals, emotions, and internally generated thought. Yet experience usually remains organized enough to continue as a lived trajectory rather than simply becoming whatever perturbation is strongest at the moment. The difficulty is that the measurements available to an experimenter do not all answer the same question. A signal can affect the organism without being consciously accessible; something can be accessible without producing the expected motivation or action; behavior can remain stable while experience changes; an experience can occur without being remembered later; and later reflection can reorganize the future without serving as a transparent record of the original episode. Grey Matter is a theory-neutral, phenomenology-facing perturbation-response architecture for handling this problem. It requires a study to declare what changed, which property of the lived trajectory is being claimed, what relation each measurement actually establishes, and what bridge supports any inference to phenomenality. Boundedness, stability, and coherence are treated as open candidate properties rather than fixed dimensions. Maintenance or termination, selective calibration, retention, and later reflection are possible downstream relations, not definitions of phenomenality. Mature theories of attention, control, learning, memory, predictive processing, active inference, source monitoring, and neural state-space dynamics retain explanatory priority. Grey Matter is useful only where its relation typing and trajectory-level organization change what an observation licenses, what an experiment should measure, or when a higher-level interpretation should be abandoned.

1. The human problem: experience continues while everything keeps changing

Conscious experience is not produced in a quiet system. A person can be reading while bodily sensations shift, background sounds change, a memory becomes active, a new goal appears, an emotional reaction begins, and an unrelated thought competes for attention. Some of those changes enter experience strongly. Some are noticed and then disappear. Some reorganize the ongoing train of thought. Some alter behavior or physiology without clearly becoming part of what is lived. Despite this continuous traffic, experience usually retains enough organization to continue as a trajectory rather than being reset by every available signal.

That ordinary fact motivates the Containment branch. The originating intuition was roughly, "What keeps conscious experience contained or coherent?" The final scientific question is narrower and less mechanistic. It does not assume that a hidden gate, filter, wall, or controller is doing the containing. It asks how to describe and measure what happened to an ongoing lived trajectory when the organism was perturbed.

This distinction matters because outward continuity is not the same thing as experiential continuity. A person can continue doing the same task while the task feels radically more effortful, fragmented, intrusive, or discontinuous. The reverse can also occur: a person may change behavior even though the change was driven by a specific decision, motor limitation, incentive, or environmental constraint rather than by a breakdown of the lived trajectory. If the scientific claim concerns experience, the endpoint cannot silently become the definition of experience.

Grey Matter therefore treats the lived trajectory as the target whenever the claim is about what was phenomenally present over time. The framework is not a new theory of consciousness and does not solve why neural or cognitive activity is accompanied by experience. It is a measurement and identification architecture for keeping the target separate from the convenient channels through which researchers observe it.

2. The visible outcome does not tell us what happened to lived experience

Psychology and neuroscience already distinguish several relations that ordinary language often compresses into phrases such as "the person had the information" or "the person knew." A stimulus can be physically available to the organism, encoded or discriminated by some system, accessible for report, phenomenally experienced, affectively significant, motivationally recruiting, remembered later, control-relevant, physiologically effective, or behaviorally expressed. Those relations interact, but one does not automatically stand in for the others (Block, 1995; Seth et al., 2008; Tsuchiya et al., 2015).

Established dissociations make the point concrete. Masked information can influence later processing without meeting ordinary criteria for conscious access (Dehaene et al., 1998; Naccache & Dehaene, 2001). Reward research separates hedonic impact from incentive motivation - liking is not the same relation as wanting (Berridge & Robinson, 2016). Action-selection and control theories distinguish valuation, allocation, selection, and execution (Cisek, 2007; Shenhav et al., 2013). Automaticity research shows that organized behavior can proceed without continuous conscious supervision (Moors & De Houwer, 2006).

The measurement problem in consciousness is likewise established rather than a Grey Matter discovery. Different behavioral, report-based, and neurophysiological measures carry different assumptions and can yield different classifications; first-person reports also require an explicit account of how the report is being related to the target phenomenal state (Seth et al., 2008; Gamez, 2014). Grey Matter inherits that problem and uses it as a hard constraint: a measure licenses only the relation it actually supports.

The same caution applies across time. An experience can occur without leaving an accessible later memory, and a later recollection can contain reconstruction, reinterpretation, or information generated during retrieval. Memory is therefore evidence about a later relation to the episode, not a transparent copy of what the original episode was like. Likewise, later reflection can change future expectations or associations without retroactively becoming a measurement of the original phenomenology (McGaugh, 2015; Kensinger & Ford, 2020).

3. Five relations that Grey Matter keeps separate

The formal manuscript uses a small set of labels as bookkeeping devices. The symbols are not proposed as new psychological mechanisms. Their job is to stop a demonstrated relation from quietly turning into a stronger one.

Scientific question	Relation	Label	Does not by itself establish
Could x causally reach the organism?	Causal exposure	E_x	Encoding, access, phenomenality, or a downstream effect
Did some organismic state encode or respond to x?	Organismic encoding or sensitivity	R_x	Conscious access or phenomenal presence
Is there operational evidence that x is consciously accessible?	Conscious access	A_x	Identity with phenomenality
Was x phenomenally present?	Lived	plain-language target	Memory, action, control, persistence, or any specific later consequence
What happened afterward?	Named consequence-specific outcome	C_x	Proof that the original state was lived

These labels matter only where they preserve a distinction. The companion will mostly use the ordinary-language relations after this point. The key rule is that no row can be substituted for another without an independent bridge.

4. The lived trajectory is the target when the claim is about experience

The most important term in the architecture is deliberately the least technical: lived. It means phenomenal presence - there is something it is like for the organism to undergo the state. Lived is not defined by later memory, successful report, voluntary control, overt action, physiological change, persistence, reward, or any other downstream consequence. That exclusion is methodological. If a state counted as lived only when it changed later behavior or memory, then experiments asking whether lived experience has those consequences would be circular.

Excluding downstream efficacy from the definition does not imply that phenomenality is causally inert. Once a study has an independently defended bridge to phenomenal presence, the question of what that experience subsequently contributes becomes empirical. It may contribute to maintaining the current state, terminating it, learning, memory, interpretation, or action; it may also leave little detectable downstream consequence. A null effect does not revoke phenomenality, and phenomenal presence does not guarantee a particular effect.

This leads to the central identification rule of Grey Matter. A containment claim is supported only when a declared perturbation is linked to a declared property of the lived trajectory using evidence typed to the relation that actually bears on that property. Stability in behavior, report, physiology, memory, affect, motivation, or control cannot substitute for stability of the lived trajectory unless an independently justified bridge is supplied.

The rule is intentionally demanding but not maximalist. A study does not need to measure every possible relation. It needs to measure the relations required by the inference it actually makes, and it needs to say what unmeasured relations are not being inferred. The framework therefore aims to narrow claims, not to require an impossible complete description of the organism.

5. What "containment" means after the mechanism claims are removed

Containment names a stabilization and organization problem. In the authoritative manuscript it is defined as the distributed and potentially mechanistically heterogeneous processes by which conscious experience remains sufficiently bounded, stable, and coherent despite ongoing changes in internal and external information. The definition does not identify a physical container, a single anatomical gate, deliberate suppression, or a hidden causal force.

The word "sufficiently" matters. Grey Matter does not treat stronger constraint as universally better. A lived trajectory can be too susceptible to irrelevant perturbation, but it can also be too rigid to incorporate information that should change it. The target is a viable dynamical range under a declared person or organism, task, state, and timescale. Adaptiveness is therefore an outcome judgment, not part of the definition.

The framework also rejects a single adjustable threshold. History can change how particular cues, contexts, or transition classes influence later experience. One cue can acquire less power to redirect the trajectory while another acquires more because it proves predictive or consequential. A signal can remain fully experienced while its influence over the next transition changes. Established learning and attention mechanisms should be used to explain those changes when they suffice (Anderson, 2013; Dunsmoor & LaBar, 2013).

Boundedness, stability, and coherence are therefore candidate properties of the lived trajectory rather than three guaranteed dimensions. Their conceptual separation is useful only if measurement can earn it.

Candidate property	What it asks	Possible indicators	Required reduction if it fails
Boundedness	How selectively perturbations are incorporated into, persist within, or reorganize the lived trajectory.	Interruption susceptibility; incorporation probability; redirection or termination by a declared perturbation.	If it cannot dissociate from stability or coherence, reduce the dimensionality.
Stability	Persistence, resistance to displacement, and recovery across time. Stability is not stasis.	Dwell time; recovery after interruption; transition rate; subjective discontinuity.	If the apparent effect is fully explained by another construct or cannot be separated from the other candidate properties, collapse it.
Coherence	Structured mutual constraint among experiential contents and transitions. Not truth, calmness, normality, or task success.	Source/context consistency; preservation of task-model relations; structured continuity reports.	If it cannot be separately measured or predicted, merge it with the lower-dimensional structure supported by data.

The pairwise dissociation logic is deliberately provisional. A trajectory might readily incorporate many events yet remain organized around a persistent theme; or it might resist external capture while its internally generated contents shift rapidly. These are candidate contrasts, not established syndromes. If sensitive studies cannot separate them, the architecture should become lower-dimensional rather than protecting the labels by definition.

6. A perturbation must be described before its effect is interpreted

A perturbation is a declared change with the potential to alter the current lived trajectory. Grey Matter does not assume a universal perturbation score. Instead it asks which features of the change are relevant to the specific comparison. Candidate properties include endpoint displacement or magnitude, rate of change, duration or temporal extent, explicit predictability, history-relative familiarity, source or information channel, and the state of the receiving organism.

These variables are not all asserted to be independently necessary. Rate matters only if it explains something that endpoint magnitude, surprise, arousal, attentional capture, or another established variable does not. Familiarity matters only if it contributes beyond explicit predictability or ordinary learning strength. Source is useful only if matched comparisons reveal something beyond the local mechanism recruited by the different channels. Fatigue, arousal, task load, goals, and pharmacological state are moderators of response, not intrinsic properties of an external event.

This separation creates experiments that can fail cleanly. A rate test can hold the starting state, final displacement, and forewarning approximately constant while changing how quickly the transition occurs. A familiarity test can match explicit predictive information while comparing a history-familiar transition with a structurally similar novel one. If the added variable contributes no reproducible information after established alternatives are modeled, the variable should be removed from the scaffold.

The same discipline applies to the trajectory side. A claim about boundedness should not be tested only with a stability measure, and a claim about coherence should not be inferred from calm behavior. Measures can cross-load and require theory, which is precisely why the target property and bridge must be declared before interpreting the result.

7. Why the architecture is branching rather than a serial pipeline

The early Containment work explored a more serial picture in which information might move from existence to representation, weighting, authority, and consequence. That formulation did not survive. Established science requires bypasses and dissociations. Information can influence the organism without ordinary conscious access. Conscious availability can occur without a particular motivational or behavioral consequence. Behavior can be organized automatically. A lived episode can be retained weakly or not at all. Later reflection is a new episode, not simply another stage of the original one.

Grey Matter therefore does not ask which single checkpoint information passed. It asks which relations have actually been demonstrated and which mechanisms are plausible for the observed pattern. Distributed competition, state-dependent precision or gain, attentional allocation, learned routing, automaticity, control allocation, action selection, and source-monitoring processes already provide non-homuncular ways for some information to exert more influence than other information (Cisek, 2007; Feldman & Friston, 2010; Moors & De Houwer, 2006; Shenhav et al., 2013).

This is why terms such as gate, filter, permission, or breakwater remain metaphors at most. If an experiment shows that a perturbation was accessible but had little motivational effect, that is a relation pattern. Claiming that a defensive process inhibited the information is a separate mechanistic hypothesis. Claiming that learned automatic organization reduced the need for conscious supervision is another. Grey Matter does not allow the higher-level label to replace those mechanism-specific explanations.

8. Four things that can happen after an experience - without defining the experience by them

The final Containment manuscript restores a question that can easily be lost when phenomenality and consequence are kept separate: once something is lived, can that phenomenal state contribute to what happens next? The framework treats four downstream relations as optional and analytically separable.

Maintenance or termination is the first. Overt change is not the only possible outcome. A person can continue a task, remain in an interaction, preserve a quiet valued state, or refrain from an immediately available response. Response-inhibition and control-allocation models already explain many such cases and should receive explanatory priority (Aron et al., 2014; Shenhav et al., 2013). Grey Matter adds only the measurement requirement: phenomenal state, continuation or termination, control demand, and overt action should not be treated as the same variable.

Selective calibration is the second. Continued sampling can sometimes support learning about which cues really predict an outcome, which merely resemble them, and which contexts retrieve the learned relation. Fear-learning, inhibitory-learning, and discrimination literatures already provide mature mechanisms for such changes (Bouton & Moody, 2004; Craske et al., 2014, 2022; Dunsmoor & LaBar, 2013). The Grey Matter claim is not that remaining in discomfort is inherently beneficial. It is that any claim about future "containment" should specify which cue-to-trajectory relation changed and should disappear when ordinary learning models fully explain it.

Retention or future accessibility is the third. Some aspects of a lived episode can become easier or harder to retrieve later through established encoding and consolidation processes. Emotional arousal and reward motivation can influence memory (McGaugh, 2015; Miendlarzewska et al., 2016). That does not make memory a criterion for having experienced the episode. A person can live something and later fail to retrieve it, just as later accessibility can be shaped by processes that occur after the original phenomenology.

Reflection-mediated reorganization is the fourth. A later act of remembering, reinterpreting, or reappraising an episode can change future associations, expectations, emotional response, or retrieval structure (Ochsner & Gross, 2005; Chan et al., 2018; Kensinger & Ford, 2020). But the reflection is itself a later causal episode. Evidence generated during reflection cannot be used as if it were a direct measurement of the original experience.

These four branches preserve the human intuition that experience can matter without turning that intuition into a circular definition. Phenomenal discomfort does not logically entail leaving. Phenomenal value does not have to appear as overt action. Later memory does not certify original phenomenality. Reflection can change the future without perfectly revealing the past.

9. How a concrete Grey Matter study should work

The architecture becomes scientific only when it constrains design. A concrete study starts with a declared perturbation and a declared property of the lived trajectory. It then specifies which relation each measurement establishes, what bridge supports any claim about phenomenality, which mature mechanism-specific models compete with the higher-level interpretation, and what result would cause the containment claim to narrow or disappear.

One basic design measures several relations to the same referent in the same participant and state. Depending on the question, this might include immediate report, affective response, motivation, memory, source judgment, control allocation, maintenance or termination, physiology, and overt behavior. The goal is not to average them into a single containment score. The goal is to learn whether the relation needed for the trajectory claim moves with or apart from the convenient downstream measures.

Another design holds one outward relation approximately constant while testing another. Report accuracy can be matched while affect, motivation, memory, or action changes. Overt behavior can be matched while the amount of conscious monitoring, conflict, or experiential disruption differs. These designs do not prove a proprietary containment mechanism. They reveal when a proxy is insufficient for the inference a study wants to make.

Within-person transitions are especially useful because the same nominal perturbation can arrive in different receiving states. Fatigue, learning, task load, or another declared moderator can be varied while local theories are modeled first. Repeated exposure can test whether change is cue-specific, context-specific, or merely a global drop in response. If the local learning, vigilance, or control model fully explains the effect, that model wins.

Intervention specificity provides another safeguard. An attentional manipulation need not change motivation; a motivational manipulation need not change episodic memory; a source-monitoring manipulation need not change behavioral vigor. If a mature mechanism predicts the selective effect, Grey Matter should not rename the effect. A

higher-level role is earned only if the trajectory analysis prevents a wrong inference, reveals a reproducible cross-mechanism regularity, or adds prediction beyond the competing account.

The methods needed for this program are not new. Dissociation logic, within-person designs, factorial comparisons, multivariate measurement, model comparison, and process-oriented measurement are established tools (Jacoby, 1991; Shamay-Tsoory et al., 2009). Grey Matter is a proposal about how to organize them around the lived trajectory without confusing the measurement channel with the target.

10. What established theories already explain

The scientific value of Grey Matter depends on resisting the temptation to absorb mature explanations into one umbrella term. Dynamic-thought work already treats mental contents and transitions as constrained processes (Christoff et al., 2016), and work on stability and variability of conscious thought already addresses how deliberate and automatic constraints shape those dynamics (Zamani et al., 2022). Experience-sampling results show that distinguishable first-person dimensions of thought dynamics can be measured without establishing Grey Matter's particular candidate properties (Sheth et al., 2026).

At other levels, allostasis precedents stability through change (McEwen, 1998); interoception, internal and external attention, and self-generated thought already organize differences in information source and coupling (Quadt et al., 2018; Chun et al., 2011; Smallwood & Schooler, 2015). Neural state-space approaches can describe restricted regions, transition structure, accessibility, and stability of neural states associated with conscious experience (Smith, 2026). Active-inference accounts can jointly explain prediction, precision, action, learning, and candidate mechanisms of conscious organization (Laukkonen et al., 2025; Whyte et al., 2026).

None of these literatures is being redescribed as Grey Matter. Their priority is a core part of the architecture. If arousal, attentional capture, prediction error, precision or volatility, working-memory interference, control demand, sensory gating, source monitoring, automaticity, learning, memory, active inference, or a neural state-space model fully explains the declared perturbation-to-trajectory effect, the mature construct should be used and the higher-level containment label should be dropped for that effect.

The residual proposal is therefore organizational and inferential. It couples a declared perturbation and a declared lived-trajectory property to relation-typed evidence, an explicit phenomenality bridge where needed, optional downstream relations, mature competing explanations, and rules for loss. The general measurement problem is precedented (Seth et al., 2008; Gamez, 2014). The question left open is whether putting those requirements together at the perturbation-to-lived-trajectory level changes experimental design or interpretation enough to justify the architecture.

11. How Grey Matter must shrink, collapse, or disappear

Grey Matter is intentionally self-pruning. Its loss conditions are part of the theory rather than ceremonial caveats. A skeptical reader should be able to identify a result that removes a variable, collapses a distinction, hands an effect back to a mature theory, or makes the higher-level architecture unnecessary.

Loss condition	What makes Grey Matter give way
Dimensional collapse	If boundedness, stability, and coherence do not dissociate under sensitive manipulation, reduce them to the lower-dimensional structure the data support.
Prior-art / subsumption collapse	If a mature dynamic-thought, active-inference, state-space, or other framework already supplies the same perturbation-to-lived-trajectory identification architecture, narrow or abandon the residual Grey Matter contribution.
Lower-dimensional victory	If arousal, salience, surprise, precision/volatility, control demand, source monitoring, sensory gating, automaticity, or another mature construct fully explains the declared effect, use that construct and drop containment for that effect.
Relation-audit redundancy	If relation typing never changes an inference, prediction, experimental design, or interpretation beyond ordinary good practice, the non-entailment layer adds no residual scientific value.

Loss condition	What makes Grey Matter give way
Perturbation redundancy	If rate, duration, familiarity, source, or another proposed perturbation variable adds no reproducible information beyond endpoint magnitude, predictability, or established alternatives, remove it.
Phenomenology failure	If the measurements cannot support a defensible bridge to the lived trajectory, report only what was actually measured and do not claim containment of experience.
Post-hoc rescue failure	If containment is invoked only after the result appears and can explain any direction of change, abandon the construct as unfalsifiable.
Causal-role redundancy	If maintenance or termination is fully explained by response inhibition, avoidance, reward, action selection, or control-allocation models and relation typing adds nothing, keep the mature explanation and drop the higher-level role.
Calibration reduction	If repeated experience is fully captured by habituation, inhibitory learning, prediction error, safety learning, discrimination, reinforcement learning, or another mature account, those mechanisms win.
Retention reduction	If later accessibility is fully explained by established encoding, consolidation, retrieval, reward-memory, or emotional-memory mechanisms, describe it in those terms rather than as a Grey Matter-specific process.
Reflection null	If later reflection changes only report or narrative and produces no reproducible future change in perturbation response, expectation, association, or retrieval structure, narrow the claim to the effects actually measured.
Episode-separation failure	If a design cannot distinguish evidence about the original lived episode from evidence generated during later recollection or reflection, it cannot use the later measure to reconstruct the original phenomenology or causal role.

12. Scope, safety, and what the framework does not claim

Grey Matter does not solve the hard problem, identify a neural seat or generator of consciousness, discover a universal containment mechanism, or prove that boundedness, stability, and coherence are independent dimensions. It does not infer stable experience from stable behavior, infer phenomenality from physiology, identify memory with prior experience, or infer that a lived state caused later action simply because the action followed it.

The framework is also not a generic theory of resilience, suppression, coping, emotional regulation, or clinical adaptation. The discussion of maintained discomfort and inhibitory learning is not advice to remain in harmful, unsafe, coercive, or unwanted situations. Termination can itself be the appropriate controlled outcome. Likewise, greater stability is not automatically healthier, and greater openness to perturbation is not automatically a failure.

The word containment and the breakwater image are pedagogical tools. They should disappear whenever they tempt the reader to imagine a literal wall or inner gate. Grey Matter is branding, not an additional mechanism. The stable scientific commitment is the identification discipline: declare the perturbation, declare the lived-trajectory property, type the evidence by the relation it measures, justify the bridge to phenomenality, compare mature alternatives, and state the conditions under which the higher-level interpretation loses.

13. Conclusion

The ordinary problem behind Containment is easy to recognize: a conscious person is constantly changing and being changed, yet experience usually remains organized enough to continue. The scientific difficulty is that the measurements available to us slice that process in different ways. Something can affect the organism without being lived. Something can be lived without controlling the next action. Behavior can remain stable while experience changes. Memory can fail after a genuine experience, and later reflection can change the future without perfectly revealing the past.

Grey Matter does not explain those patterns by adding a hidden container. It turns them into an identification problem. What changed? Which property of the lived trajectory is actually being claimed? What relation does the measurement establish? What bridge connects that relation to phenomenality? Which established theory already explains the effect? What would make the higher-level interpretation disappear?

Boundedness, stability, coherence, perturbation rate, familiarity, source, and downstream branches are retained only while they earn independent scientific work. Phenomenal presence remains separate from maintenance, learning, retention, and reflection. Mature theories keep their own mechanisms. The architecture succeeds only if this discipline changes what researchers measure, compare, or conclude. If it does not, Grey Matter should collapse back into the established science from which it was assembled.

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